Illiquidity and Stock Returns: Cross-Section and Time-Series Effects

Yakov Amihud

AUTHORS	Yakov Amihud, Stern School of Business, New York University
PUBLICATION YEAR	2002
JOURNAL	Journal of Financial Markets, Vol. 5, No. 1, pp. 31–56
DOI	10.1016/S1386-4181(01)00024-6
SOURCE	https://www.cis.upenn.edu/~mkearns/finread/amihud.pdf
RESEARCH AREA	Asset Pricing under Market Frictions; Market Microstructure
SERIES	Market Microstructure
PAPER TYPE	Empirical, with a light theoretical scaffold (cross-sectional and time-series asset pricing)
REVIEWER	Arthur Faugeron
DATE REVIEWED	2 September 2026

ABSTRACT

This paper shows that, over time, expected market illiquidity positively affects the ex ante stock excess return, suggesting that expected stock excess return partly represents an illiquidity premium. This complements the well-established cross-sectional positive relationship between illiquidity and stock return. Stock returns are also shown to be negatively related, over time, to contemporaneous unexpected illiquidity. The illiquidity measure employed, denoted ILLIQ, is the average across stocks of the daily ratio of absolute stock return to dollar trading volume, a quantity easily constructed from ordinary daily return and volume data and therefore available for long time series in most stock markets, unlike bid-ask spreads or transaction-level microstructure measures. Using NYSE data for 1963–1997, the paper documents a positive and significant cross-sectional illiquidity premium, and, using annual and monthly aggregate illiquidity series, a positive effect of expected market illiquidity on ex ante excess return and a negative effect of unexpected illiquidity on contemporaneous excess return. Both effects are markedly stronger for small-capitalization stocks, which the paper argues helps explain time variation in the small-firm premium.

1. Introduction

Amihud (2002) sits at the point where two literatures meet: the cross-sectional asset-pricing literature that, since **Amihud and Mendelson (1986)**, treats illiquidity as a priced characteristic, and the time-series asset-pricing literature that treats the equity premium as a compensation for risk that varies predictably over time (French, Schwert and Stambaugh, 1987; Fama and French, 1989). The paper's contribution is to ask whether the same illiquidity channel that operates across stocks at a point in time also operates across time for the market as a whole, and whether it can be measured with data that are available for almost any stock market, in contrast to the bid-ask spread and transaction-level microstructure measures used in earlier illiquidity studies.

The vehicle for this is a new, deliberately coarse illiquidity measure, *ILLIQ*, defined as the average daily ratio of a stock's absolute return to its dollar trading volume. The measure requires only daily return and volume data, which are available for NYSE stocks back to 1963 and, more generally, for most exchanges over long spans of time, unlike quoted or effective bid-ask spreads, Kyle's lambda, or the probability of informed trading, all of which require intraday transactions-and-quotes data that exist only for recent, often short, sample periods. This tradeoff between precision and time-series length is the paper's central methodological wager: a noisier proxy is accepted in exchange for the ability to run 34 years of annual and monthly time-series tests that finer measures cannot support.

Why the question matters. If the equity premium partly compensates investors for the cost of trading out of a position, rather than purely for holding risk, then the standard interpretation of the historical equity premium as a pure risk price is incomplete, and quantities that are usually treated as risk-only (the market excess return, the small-firm premium) must be reinterpreted as compounds of a risk price and a liquidity price. This bears directly on the equity premium puzzle: part of the historically high equity premium may reflect the fact that stocks are less liquid than the Treasury bills against which the premium is computed, not that investors are more risk averse than standard models assume.

1.1 What Was Already Known

Before this paper, the illiquidity-return relationship had been established almost exclusively in the cross-section: stocks with wider bid-ask spreads (Amihud and Mendelson, 1986; Eleswarapu, 1997), higher amortized effective spreads (Chalmers and Kadlec, 1998), larger price impact of order flow (Brennan and Subrahmanyam, 1996), or a higher probability of informed trading (Easley, Hvidkjaer and O'Hara, 1999 draft) earn higher risk-adjusted average returns. Separately, the time-series asset-pricing literature had shown that ex ante equity premia vary predictably with proxies for risk and business conditions — volatility (French, Schwert and Stambaugh, 1987), the default and term yield premiums on corporate and Treasury bonds (Fama and French, 1989; Fama, 1990; Keim and Stambaugh, 1986). What was missing was a bridge between the two literatures: no study had tested whether the same illiquidity channel that prices stocks in the cross-section also drives time variation in the aggregate equity premium, largely because the microstructure-based illiquidity measures used in the cross-sectional literature simply do not exist as long clean time series.

1.2 Contribution

The paper's contribution is threefold: (i) it proposes and validates *ILLIQ* as a volume-based illiquidity proxy that correlates strongly with microstructure-based measures (Brennan and Subrahmanyam's price-impact and fixed-cost estimates) while requiring only daily data; (ii) it shows, using this proxy, that expected market illiquidity forecasts the market excess return over the following year or month, and that innovations in market illiquidity move contemporaneous excess returns in the opposite direction; and (iii) it shows that both effects are substantially stronger for small-capitalization stocks, offering an illiquidity-based explanation for why the small-firm premium itself varies over time. CRITIQUE The claimed contribution is real but narrower than "illiquidity is priced": what the time-series tests actually establish is a predictive correlation between a lagged illiquidity proxy and subsequent market excess return, filtered through a specific autoregressive forecasting model. Section 7 evaluates how much weight that correlation can bear.

2. Research Problem and Hypothesis

2.1 Research Question

The paper asks two linked questions. First, cross-sectionally: does a stock's illiquidity, measured by the average ratio of its absolute daily return to its dollar trading volume, command a positive risk-adjusted return premium on the NYSE after controlling for beta, size, dividend yield and past returns? Second, and this is the paper's genuine novelty: does *expected* market-wide illiquidity, estimated from an autoregressive forecasting model, positively predict the market's ex ante excess return over Treasury bills, and does *unexpected* market illiquidity negatively affect the contemporaneous market return?

2.2 Hypotheses

Three hypotheses are stated formally and tested in sequence:

- **H-1** (time-series, expected illiquidity): the coefficient on lagged expected market illiquidity in a regression of market excess return is positive ($g_1 > 0$).
- **H-2** (time-series, unexpected illiquidity): the coefficient on the innovation (unexpected component) of market illiquidity is negative ($g_2 < 0$).
- **SZ-1 / SZ-2** (size cross-cut): both the expected-illiquidity coefficient and the (negative) unexpected-illiquidity coefficient decline monotonically in absolute value as firm size rises across size-decile portfolios — the illiquidity channel is strongest for small stocks and weakest for large stocks.

Underlying all three is the economic proposition that the market excess return, conventionally interpreted purely as a risk premium, is in part a compensation demanded by investors for holding a less liquid asset class than Treasury bills, so that time variation in expected illiquidity should show up as time variation in the required excess return.

3. Theoretical Framework and Mechanism

The paper is not a fully specified equilibrium model; it borrows the microstructure logic of Kyle (1985) and Amihud and Mendelson (1986) to motivate a reduced-form measure and then builds a purely statistical time-series forecasting apparatus around it. The mechanism has three layers: (a) how illiquidity is defined and measured, (b) how it enters a cross-sectional pricing regression, and (c) how expectations of it are formed and fed into a time-series pricing regression.

3.1 The Illiquidity Measure

Illiquidity is defined as the price impact per dollar of order flow. Under Kyle's (1985) framework, a market maker who cannot distinguish informed from noise order flow sets prices as an increasing function of order imbalance; the resulting price sensitivity to volume, λ, is a natural illiquidity concept, but estimating it requires transaction-by-transaction data. Amihud constructs a coarse empirical analogue for stock i in year y:

$$ILLIQ_{iy} = (1/D_{iy}) \sum_{t=1}^{D_{iy}} |R_{iyd}| / VOLD_{iyd} \tag{1}$$

R_{iyd} return on stock i on day d of year y.

$VOLD_{iyd}$ dollar trading volume of stock i on day d of year y.

D_{iy} number of trading days with available data for stock i in year y.

Equation (1) is a within-stock average of the ratio of absolute return to dollar volume. Economically, it reads as the percentage price move induced by one dollar of trading — a coarse, low-frequency substitute for Kyle's lambda that trades statistical precision (it is a daily-average ratio of two noisy quantities, not an estimated slope coefficient from a properly specified impact regression) for availability, since it needs only close-to-close returns and daily volume, both routinely available even where quotes and transaction records are not. The assumption required for equation (1) to be a valid illiquidity proxy is that a given day's absolute return is predominantly driven by the volume needed to accommodate it — not by public information arriving without any associated trade, and not by volume unrelated to price-moving order flow (e.g., pure liquidity or index-rebalancing volume that leaves price unmoved). Both violations bias $ILLIQ$ in ways the paper does not formally address (see Section 8).

Aggregate market illiquidity in year y is the cross-sectional average:

$$AILLIQ_y = (1/N_y) \sum_{i=1}^{N_y} ILLIQ_{iy} \tag{3}$$

and, because $AILLIQ_y$ varies greatly in level across the 34 sample years, the cross-sectional regressions instead use a mean-adjusted, dimensionless version of the individual-stock measure:

$$ILLIQMA_{iy} = ILLIQ_{iy} / AILLIQ_y \tag{4}$$

which normalizes each stock's illiquidity by that year's cross-sectional average, so the mean of *ILLIQMA* is 1 in every year and the cross-sectional coefficient on it is comparable across years despite trending nominal illiquidity levels (nominal *ILLIQ* levels are not stationary because dollar volumes grow over decades while price levels also change, so without normalization a raw pooled cross-sectional regression across 34 years would implicitly weight years unevenly).

3.2 Cross-Sectional Pricing Equation

The paper follows the Fama and MacBeth (1973) two-step procedure. In each of 408 months (1964–1997), stock returns are regressed cross-sectionally on lagged characteristics:

$$R_{imy} = \boldsymbol{\kappa}_{0my} + \sum_{j=1}^{J} \boldsymbol{\kappa}_{jmy} X_{ji,y-1} + u_{imy} \tag{2}$$

R_{imy} — return on stock i in month m of year y, delisting-adjusted following Shumway (1997) to avoid survivorship bias.

$X_{ji,y-1}$ — characteristic j of stock i, estimated from data in year $y-1$ and hence known to investors at the start of year y.

$\boldsymbol{\kappa}_{jmy}$ — month-specific price of characteristic j; averaged over the 408 months and t-tested against zero.

The included characteristics are *ILLIQMA*, *BETA* (estimated on ten size-sorted portfolios by the Scholes-Williams 1977 method, equation (5): $R_{pty} = \alpha_{py} + BETA_{py} \cdot RM_{ty} + \varepsilon_{pty}$), the log of market capitalization ($\ln\ SIZE$), the standard deviation of daily returns (*SDRET*), the dividend yield (*DIVYLD*), and two past-return variables, *R100* and *R100YR*. The prediction is $\boldsymbol{\kappa}_{ILLIQ} > 0$: a stock's expected return rises in its lagged illiquidity, holding risk and size fixed. This is the standard Fama-MacBeth machinery; the theoretical content is entirely inherited from Amihud and Mendelson (1986) rather than derived anew here.

3.3 The Time-Series Mechanism — The Paper's Central Contribution

The time-series apparatus is the part of the paper that is genuinely new, and it is built in four explicit steps, each with its own equation.

STEP 1 — THE TARGET RELATIONSHIP

$$E[(R_M - R_f)_y \mid \ln AILLIQ^E_y] = \boldsymbol{\varphi}_0 + \boldsymbol{\varphi}_1 \ln AILLIQ^E_y \tag{6}$$

where R_{My} is the equal-weighted NYSE market return, R_{fy} is the one-year Treasury bill yield, and $\ln AILLIQ^E_y$ is expected market illiquidity for year y formed on information available in year $y-1$. The hypothesis is $\varphi_1 > 0$: this is the direct statement of "illiquidity is priced over time," but equation (6) is not estimable as written because $AILLIQ^E_y$ is unobserved — it must first be modeled.

STEP 2 — FORMING THE EXPECTATION

Market illiquidity is assumed to follow a first-order autoregressive process:

$$ln\ AILLIQ_y = c_0 + c_1\, ln\ AILLIQ_{y-1} + v_y \tag{7}$$

with the expectation formed mechanically as the model's fitted value:

$$ln\ AILLIQ^E{}_y = c_0 + c_1\, ln\ AILLIQ_{y-1} \tag{8}$$

Investors are thus modeled as naive AR(1) forecasters of the log illiquidity series, with the residual v_y interpreted as the unexpected (innovation) component of illiquidity. Because OLS estimates of an AR(1) coefficient are biased downward in finite samples, the paper applies Kendall's (1954) bias-correction, adding $(1+3d_1)/T$ to the raw OLS estimate d_1. Estimated on annual data, equation (7) yields $ln\ AILLIQ_y = -0.200 + 0.768\ ln\ AILLIQ_{y-1}$ (t = 1.70, 5.89; R^2 = 0.53), with the bias-corrected slope rising to 0.869 — a persistence coefficient close to but below one, consistent with a highly persistent but mean-reverting illiquidity process.

STEP 3 — SUBSTITUTING INTO THE RETURN EQUATION

Substituting (8) into (6) collapses the two-step forecasting model into a single reduced-form regression on the lagged, observed illiquidity level:

$$(R_M - R_f)_y = \varphi_0 + \varphi_1\, ln\ AILLIQ^E{}_y + u_y = g_0 + g_1\, ln\ AILLIQ_{y-1} + u_y \tag{9}$$

with $g_0 = \varphi_0 + \varphi_1 c_0$ and $g_1 = \varphi_1 c_1$. This is the key algebraic move in the paper: because expected illiquidity is itself a linear function of lagged realized illiquidity, testing "does expected illiquidity forecast returns" reduces to testing whether the market excess return can be predicted from last year's illiquidity level alone — an ordinary predictive regression that does not require observing investors' expectations directly. The cost of this convenience is that the estimated g_1 conflates the true illiquidity price φ_1 with the persistence of illiquidity c_1; the two are not separately identified from (9) alone.

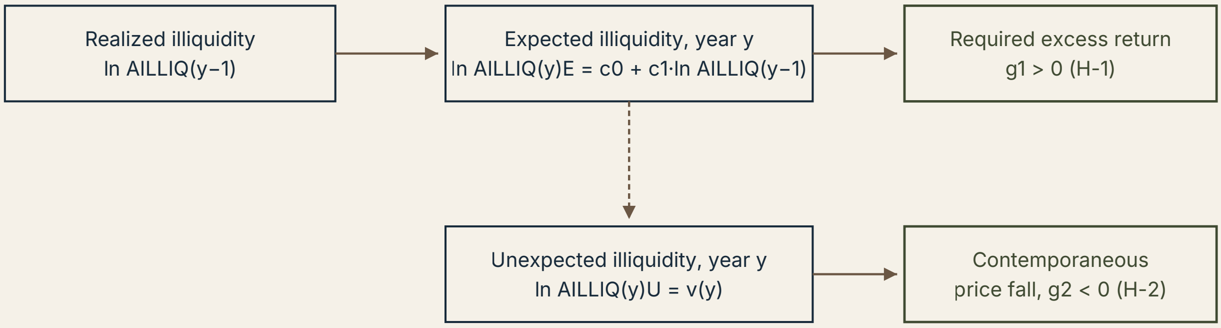
STEP 4 — ADDING THE UNEXPECTED-ILLIQUIDITY CHANNEL

The full estimated model adds the innovation term, giving the equation actually taken to the data:

$$(R_M - R_f)_y = g_0 + g_1\, ln\ AILLIQ_{y-1} + g_2\, ln\ AILLIQ^U{}_y + w_y \tag{10}$$

where $ln\ AILLIQ^U{}_y = v_y$, the bias-corrected residual from (7). Two testable predictions follow: *H-1:* $g_1 > 0$ and *H-2:* $g_2 < 0$. The economic logic behind H-2 is a feedback story: if $c_1 > 0$, an unexpectedly high realization of illiquidity this year raises the market's forecast of next year's illiquidity, which (by H-1) raises the required future return, which (holding future cash flows fixed) can only be achieved today by a fall in the current price. Unexpectedly high illiquidity should therefore coincide with a contemporaneous negative abnormal return — a discount-rate effect running in real time, distinct from the expected-illiquidity effect which operates through the ex ante required return over the coming year. The mechanism is summarized schematically in Figure 1.

Figure 1. **Schematic of the expected/unexpected illiquidity transmission mechanism**



Higher realized illiquidity raises the illiquidity forecast, which raises the required future return and, holding cash flows fixed, forces the current price down.

Author's schematic reconstruction of the causal chain implied by the paper's equations (7)–(10); not reproduced from the original article.

4. Data and Empirical Strategy

4.1 Sample and Data Sources

The sample is restricted to NYSE-listed common stocks, 1963–1997 (illiquidity and characteristics computed on 1963–1996 data, returns regressed over 1964–1997), drawn from CRSP daily and monthly files. NASDAQ and AMEX stocks are excluded explicitly to avoid mixing microstructure regimes: NASDAQ trading is dealer-intermediated, which mechanically inflates reported volume relative to the NYSE's largely direct investor-to-investor trading (Reinganum, 1990), so pooling exchanges would contaminate *ILLIQ* with a cross-exchange volume-definition artifact rather than a pure liquidity signal. Between 1,061 and 2,291 stocks satisfy the inclusion filters in a given year.

4.2 Sample Construction and Filters

- More than 200 days of return and volume data in year *y−1*, and the stock listed at the end of year *y−1*.
- End-of-year price above $5, to avoid the minimum-tick noise that dominates returns on low-priced ("penny") stocks (Harris, 1994).
- Market capitalization data available on CRSP at the end of year *y−1*, which excludes ADRs, scores and primes.
- Outlier trimming: stocks in the top or bottom 1% of the *ILLIQ* distribution in year *y−1* are dropped.

Delisting returns are handled following Shumway (1997): the last available CRSP return or the delisting return is used, and a −30% return is imputed for delistings whose reason codes correspond to unavailable, OTC-transfer, bankruptcy, or exchange-guideline failures — Shumway's estimate of the average true delisting return for such cases. **EVIDENCE** This is a meaningful survivorship-bias correction relative to older studies that simply drop delisted stocks; it is not, however, a stock-specific delisting return, so idiosyncratic delisting outcomes are still smoothed into a single imputed value.

4.3 Variable Construction

Table A. Variables in the cross-sectional model (2)

Variable	Construction	Predicted sign
ILLIQMA	Mean-adjusted illiquidity, eq. (4)	+
BETA	Scholes-Williams beta of the size-decile portfolio to which the stock belongs, eq. (5)	+
ln SIZE	Log market capitalization, year-end $y-1$	–
SDRET	Std. dev. of daily return in year $y-1$	ambiguous
DIVYLD	Annual cash dividends ÷ year-end price	ambiguous
R100	Return over the last 100 trading days of year $y-1$	+
R100YR	Return over the earlier part of year $y-1$ (start of year to 100 days before its end)	+

Source: constructed by the reviewer from the variable definitions given in Amihud (2002), Sections 2.3.1–2.3.3.

4.4 Identification Logic

The cross-sectional identification is standard Fama-MacBeth: characteristics are dated at $y-1$, strictly predetermined relative to the return in month m of year y, so the monthly cross-sectional coefficients are, in principle, free of contemporaneous simultaneity between a stock's realized return and its measured characteristics. The identifying variation is purely cross-sectional and within-month: in a given month, which stocks happen to have had higher lagged illiquidity earn higher realized return, after partialling out beta, size, dividend yield and momentum.

The time-series identification is weaker and rests on the predictive regression design of French, Schwert and Stambaugh (1987): the right-hand-side illiquidity variables in equation (10) are lagged (the AR(1)-forecasted expected component) or estimated as residuals from a first-stage regression (the unexpected component), so the regression is, by construction, a forecasting exercise rather than a structural one. **Where the identifying variation comes from** is the year-to-year (or month-to-month) fluctuation in a single scalar aggregate — average market-wide *ILLIQ* — regressed against a single scalar aggregate, the equal-weighted market excess return. With only 34 annual observations (T=34) or 408 monthly observations that are themselves overlapping draws from a persistent, highly autocorrelated process, the effective number of independent data points behind the annual-frequency result is small, a point the paper does not directly confront (see Section 7).

4.5 Estimation and Inference

Cross-sectional coefficients are estimated by OLS separately in each of 408 months and then averaged, with a t-test on the time series of monthly coefficients (Fama-MacBeth standard errors), and cross-checked with weighted least squares to address cross-sectional heteroskedasticity. Time-series regressions (10), (10sz), (10m), (11) and (11sz) are estimated by OLS with conventional t-statistics and, in the monthly and size-portfolio specifications, alternative t-statistics computed from Newey-West (1987) heteroskedasticity- and autocorrelation-consistent standard errors, or White (1980) heteroskedasticity-consistent standard errors. The paper also invokes Stambaugh's (1999) result that a predictive regression with a persistent, endogenous regressor produces an upward-biased slope estimate whenever the innovation in the regressor is correlated with the return residual (exactly the situation implied by H-2, since $Cov(u_y, v_y) < 0$ is expected); its stated remedy is to include the illiquidity innovation v_y directly in the return regression, which is precisely the structure of equation (10).

5. Results

5.1 Descriptive Statistics

Table 1. Statistics on variables (NYSE, 1963–1996)

Variable	Mean of annual means	Mean of annual S.D.	Median of annual means	Min. annual mean	Max. annual mean
ILLIQ	0.337	0.512	0.308	0.056	0.967
SIZE ($million)	792.6	1,611.5	538.3	263.1	2,195.2
DIVYLD (%)	4.14	5.48	4.16	2.43	6.68
SDRET	2.08	0.75	2.07	1.58	2.83

Source: Amihud (2002), Table 1. ILLIQ multiplied by 10^6.

The reported correlation $Corr(ILLIQ_{iy}, \ln SIZE_{iy}) = -0.614$ confirms that illiquidity and size are strongly, but not perfectly, related — leaving room for *ILLIQ* to carry information beyond size, which is exactly what the multivariate cross-sectional regressions are designed to test. The correlation between *ILLIQ* and *SDRET* is low (0.278), which the paper uses to argue that illiquidity is not simply relabeled volatility risk.

5.2 Cross-Sectional Results

Table 2 (reproduced in part below) reports Fama-MacBeth average coefficients from 408 monthly cross-sectional regressions. The full-model coefficient on *ILLIQMA* is 0.163–0.216 across specifications and subperiods, always significant at conventional levels, with t-statistics between 3.7 and 6.6. Of the 408 monthly coefficients, 63.4% are positive, a proportion the paper reports as significantly different from the 50% expected by chance. The illiquidity effect survives excluding January (mean 0.126, t = 5.30), a meaningful check given that beta and size effects in older studies were known to concentrate in January (Keim, 1983; Tinic and West, 1986).

Table 2 (excerpt). Cross-sectional regression of stock return on illiquidity and other characteristics — full model, t-statistics in parentheses

Variable	All months	Excl. January	1964–1980	1981–1997
Constant	1.922 (4.06)	1.568 (3.32)	2.074 (2.63)	1.770 (3.35)
BETA	0.217 (0.64)	0.260 (0.79)	0.297 (0.59)	0.137 (0.30)
ILLIQMA	0.112 (5.39)	0.103 (4.91)	0.135 (3.69)	0.088 (4.56)
R100	0.888 (3.70)	1.335 (6.19)	0.813 (2.33)	0.962 (2.92)
R100YR	0.359 (3.40)	0.439 (4.27)	0.324 (2.04)	0.395 (2.82)
ln SIZE	−0.134 (−3.50)	−0.073 (−2.00)	−0.217 (−3.51)	−0.051 (−1.14)
SDRET	−0.179 (−1.90)	−0.274 (−2.89)	−0.136 (−0.96)	−0.223 (−1.77)
DIVYLD	−0.048 (−3.36)	−0.063 (−4.28)	−0.075 (−2.81)	−0.021 (−2.11)

Source: Amihud (2002), Table 2, full-model columns.

EVIDENCE The dependent variable is the monthly stock return (in percent). What changes across columns is only the sample of months and years used to compute the average coefficient, not the specification — this is a stability check, not a robustness check across alternative model specifications. The coefficient that matters most is *ILLIQMA*: taking the all-months estimate of 0.112, and noting that *ILLIQMA* is a mean-adjusted ratio with a cross-sectional standard deviation on the order of the mean (Table 1's SD/mean ratio for ILLIQ is roughly 1.5), a one-unit increase in mean-adjusted illiquidity (i.e., moving from average illiquidity to twice-average illiquidity) is associated with roughly 0.11 percentage points of additional monthly return, or about 1.3 percentage points annually — economically material relative to a historical NYSE equity premium of roughly 5–7% per year, though this calculation, not performed by the author, is the reviewer's own back-of-envelope translation and should be treated as illustrative INFERENCE rather than as a number reported in the paper.

Beta loses significance once size is included, which the paper attributes to beta being estimated on size-sorted portfolios and hence collinear with size; this is a specification artifact of the portfolio-beta procedure, not evidence that market risk is unpriced. Size (*ln SIZE*) retains a negative, significant coefficient even controlling for *ILLIQMA*, indicating that size is not merely a noisy proxy for the paper's illiquidity measure — the two variables carry at least partially distinct information, consistent with the moderate (−0.614) rather than near-unit correlation reported in Table 1.

5.3 Time-Series Results — Annual Data

Estimating the AR(1) illiquidity model (7) gives $ln\ AILLIQ_y = -0.200 + 0.768\ ln\ AILLIQ_{y-1}$ (t = 1.70 and 5.89; $R^2 = 0.53$; Durbin-Watson = 1.57), bias-corrected to a slope of 0.869. The core time-series regression (10), estimated over 1964–1996, supports both H-1 and H-2: the coefficient on lagged illiquidity is positive and significant, and the coefficient on the illiquidity innovation is negative and significant (full coefficients reported in Table 3 below, market-return column).

Table 3 (excerpt). Effect of market illiquidity on excess return by size portfolio — annual data, 1964–1996 (t-statistics in parentheses; Newey-West t-statistics in brackets)

Coefficient	RM−Rf	RSZ2−Rf	RSZ4−Rf	RSZ6−Rf	RSZ8−Rf	RSZ10−Rf
Constant	14.740	19.532	17.268	14.521	12.028	4.686
ln $AILLIQ_{y-1}$	10.226 (2.68)	15.230 (3.18)	11.609 (2.52)	9.631 (2.40)	7.014 (1.98)	0.447 (0.13)
ln $AILLIQ^U_y$	−23.567 (−4.52)	−28.021 (−4.29)	−24.397 (−3.88)	−20.780 (−3.80)	−18.549 (−3.84)	−14.416 (−3.14)
R^2	0.512	0.523	0.450	0.435	0.413	0.249

Source: Amihud (2002), Table 3.

Note that RSZ2 denotes the smallest-firm decile portfolio and RSZ10 the largest. Reading across the $ln\ AILLIQ_{y-1}$ row confirms hypothesis SZ-1: the coefficient falls monotonically from 15.230 for the smallest decile to a statistically indistinguishable-from-zero 0.447 for the largest decile. The $ln\ AILLIQ^U_y$ row confirms SZ-2: the coefficient rises monotonically (becomes less negative) from −28.021 to −14.416. The market-level R^2 of 0.512 in an annual excess-return regression is a large number by the standards of return-predictability regressions (which more commonly report R^2 in the 0.02–0.10 range at annual frequency); this is itself worth flagging as unusual and is examined critically in Section 7.

Figure 2. Illiquidity coefficients across size-decile portfolios, annual model (10sz), 1964–1996

Source: constructed by the reviewer from the coefficient values reported in Amihud (2002), Table 3. Bars scaled independently within each panel; not to a common vertical axis.

5.4 Time-Series Results — Monthly Data

The monthly AR(1) is far more persistent: $ln\ MILLIQ_m = -0.313 + 0.945\ ln\ MILLIQ_{m-1}$ (t = 3.31, 58.36; R^2 = 0.89), bias-corrected to 0.954. Estimating the monthly analogue of (10), which adds a January dummy ($JANDUM_m$) to absorb the well-documented January seasonal, again confirms both H-1 ($g_1 = 0.712$, t = 2.50) and H-2 ($g_2 = -5.520$, t = 6.21), and the size-decile pattern (SZ-1, SZ-2) reappears. A six-subperiod split (68 months each) shows all six subperiod estimates of g_1 positive (mean 0.871, median 0.827) and all six of g_2 negative (mean −7.089, median −5.984), evidence of reasonable temporal stability in sign, though the paper does not report whether the subperiod point estimates are individually statistically distinguishable from each other.

5.5 Controlling for Bond Yield Premiums

Adding the default premium ($DEF_m = Y_{BAA,m} - Y_{AAA,m}$) and the term premium ($TERM_m = Y_{LONG,m} - Y_{TB3,m}$) in equation (11) leaves both illiquidity coefficients significant with the predicted signs, and both bond-market variables enter with the expected positive sign, consistent with Fama and French (1989). The default premium's coefficient itself declines monotonically with firm size across the size-decile regressions — a second, independent piece of evidence (beyond the illiquidity coefficients) that small firms are disproportionately exposed to aggregate distress-type risk factors, which strengthens confidence that the illiquidity size-gradient is not a statistical artifact specific to the illiquidity variable alone.

6. Robustness and Additional Tests

CONCERN: SURVIVORSHIP / DELISTING BIAS IN CROSS-SECTIONAL RETURNS

Test — delisting returns are imputed following Shumway (1997) rather than dropping delisted stocks. **Result —** the paper reports this as standard practice without a with/without comparison. **Does it resolve the concern?** Partially. The Shumway correction is a recognized improvement over simply excluding delisted stocks, but a single imputed −30% return for several distinct delisting reasons is a coarse fix, and the paper does not show that the illiquidity coefficient is insensitive to this choice.

CONCERN: THE ILLIQUIDITY EFFECT IS A JANUARY / SMALL-FIRM SEASONAL ARTIFACT

Test — monthly and annual regressions are re-estimated excluding January (cross-section) and with an explicit January dummy (time series, eq. 11). **Result —** the *ILLIQMA* coefficient remains positive and significant excluding January (0.126, t=5.30, vs 0.162, t=6.55 including it); the time-series g_1 and g_2 remain significant with the January dummy included. **Does it resolve the concern?** Yes for the specific January effect, which is the standard robustness check in this literature (following Keim, 1983, and others). It does not, however, rule out other calendar or momentum-driven seasonality.

CONCERN: OLS ON A HETEROSKEDASTIC, CROSS-SECTIONALLY DISPERSED PANEL UNDERSTATES CROSS-SECTIONAL STANDARD ERRORS

Test — the cross-sectional model is re-estimated by weighted least squares. **Result —** the paper states results are "qualitatively similar," with *ILLIQMA* remaining positive and significant in all subsamples, but does not report the WLS coefficient table (footnote 14: "available from the author upon request"). **Does it resolve the concern?** Only partially and on trust: an unreported robustness check that the reader cannot inspect is weaker evidence than a reported one, however plausible the author's summary.

CONCERN: THE AR(1) FORECASTING MODEL FOR ILLIQUIDITY IS UNSTABLE, SO THE "EXPECTED ILLIQUIDITY" CONSTRUCT IS NOT WELL DEFINED OVER 34 YEARS

Test — a Chow test for structural stability of the AR(1) coefficients, both annual and monthly. **Result —** the paper reports stability ("found to be stable over time, as indicated by the Chow test") without reporting the test statistic, breakpoint(s) considered, or p-value. **Does it resolve the concern?** Not fully verifiable from the text as written; the claim is asserted rather than documented with enough detail (which breakpoint, what F-statistic) for a reader to evaluate its power.

CONCERN: THE UPWARD BIAS IN A PERSISTENT PREDICTIVE REGRESSOR (STAMBAUGH, 1999) COULD BE DRIVING H-1 BY ITSELF

Test — the paper cites a simulation (unreported, footnote 19) suggesting the bias in g_1 is about 4% of its estimated value, and structurally addresses the bias by including the illiquidity innovation v_y in the same regression, per Stambaugh's proposed remedy. **Result —** g_1 remains large and significant after this correction. **Does it resolve the concern?** The structural remedy is appropriate and theoretically motivated, but the 4% bias estimate itself is not reproducible from the published paper (results "available from the author upon request"), so it functions as an assertion rather than a demonstrated robustness result.

CONCERN: ROBUSTNESS OF THE CROSS-SECTIONAL ILLIQ–MICROSTRUCTURE LINK

Test — a single cross-sectional regression of *ILLIQ* on Brennan-Subrahmanyam's Kyle-lambda and fixed-cost estimates for 1984 only: $ILLIQ_i = 292 + 247.9\lambda_i + 49.2c_i$ (t = 12.25, 13.78, 17.33; R^2 = 0.30). **Result —** strong, statistically significant validation that *ILLIQ* correlates with genuine microstructure-based illiquidity. **Does it resolve the concern?** It establishes construct validity for a single cross-section in a single year (1984); it does not establish that the relationship is stable across the full 1963–1997 sample, since microstructure-based estimates for other years are not available to check against.

7. Critical Evaluation

7.1 What the Paper Establishes

ESTABLISHED

The cross-sectional illiquidity-return relationship, replicated here with a new, volume-based measure on NYSE data spanning 1964–1997, and shown to be robust to excluding January, to controlling for beta, size, dividend yield and momentum, and to WLS re-estimation (as summarized rather than reported in full). The construct validity of *ILLIQ* as a proxy for microstructure-based illiquidity (Kyle's lambda, fixed trading costs) is also established for the single year in which a direct comparison is possible (1984).

SUPPORTED BUT CONDITIONAL

The time-series claim that expected market illiquidity forecasts the market excess return, and unexpected illiquidity moves it contemporaneously, is well supported *conditional on* the AR(1) specification for expected illiquidity being the correct model of investor expectations, and conditional on the interpretation that price movements associated with illiquidity innovations reflect a discount-rate channel rather than an unmeasured cash-flow or risk channel correlated with illiquidity.

> **SUGGESTIVE**
>
> The claim that the size effect's time variation is "explained" by illiquidity is suggestive rather than established: the paper shows that illiquidity coefficients decline monotonically with size, which is consistent with the size-effect-as-illiquidity-premium story, but it does not directly regress the size premium itself (the SMB-type spread) on illiquidity innovations and show that illiquidity accounts for a specific, quantified share of the well-known instability documented by Brown, Kleidon and Marsh (1983).

> **NOT ESTABLISHED**
>
> That the relationship is causal in the structural sense (illiquidity *causes* the required return to rise, rather than both being driven by a common latent state variable such as aggregate risk aversion or macroeconomic uncertainty) is not established and, given the observational, purely time-series nature of the evidence, cannot be established by this research design. Nor is it established that *ILLIQ* measures pure price impact rather than a mixture of price impact and return volatility, since both enter the numerator of the ratio (see Section 8).

7.2 Strengths

STRENGTHS RECOGNIZED BY THE AUTHOR

- The illiquidity measure requires only daily return and volume data, making it constructible for markets and periods where microstructure data do not exist — explicitly framed by the author as the paper's practical contribution.
- The measure correlates strongly with established microstructure-based illiquidity estimates (the 1984 Brennan-Subrahmanyam validation).

STRENGTHS IDENTIFIED INDEPENDENTLY

- The four-equation derivation from the target relationship (6) through the estimable reduced form (10) is unusually explicit about the identification chain; a reader can trace precisely how each estimated coefficient maps back to the underlying economic parameter, which is not always true of reduced-form predictive-return papers of this era.
- The size-decile cross-cut (Tables 3 and 4) functions as an internal robustness check that most predictability papers of the period lack: if the illiquidity coefficients did not decline monotonically with size, the aggregate market-level result would be much harder to credit as an illiquidity story rather than a spurious correlation, and the fact that the monotonic pattern shows up independently in both annual and monthly data, and with and without bond-yield controls, is a genuinely demanding joint test that the paper passes.
- The explicit application of Kendall's (1954) small-sample bias correction and engagement with Stambaugh's (1999) predictive-regression bias critique show methodological awareness that is ahead of much contemporaneous applied work using simple OLS predictive regressions.

7.3 Weaknesses and Critique

PROBLEM: SMALL EFFECTIVE SAMPLE SIZE AT ANNUAL FREQUENCY

Why it matters — the headline time-series result (10) is estimated on 33 annual observations (1964–1996). With two right-hand-side illiquidity variables and highly persistent regressors, the asymptotic justification for OLS t-statistics is stretched thin, and an R^2 of 0.51 for an aggregate annual excess-return regression is far outside the typical range for this literature. Potential bias — small-sample distortions in both point estimates and standard errors, in a direction that is hard to sign definitively without a dedicated small-sample simulation (which the paper does not report in full). How it could be tested — a bootstrap or Monte Carlo simulation calibrated to the estimated persistence of illiquidity, in the spirit of Stambaugh (1999) or Nelson and Kim (1993), reporting a small-sample-adjusted confidence interval for g_1 and g_2.

PROBLEM: ILLIQ CONFLATES PRICE IMPACT WITH RETURN VOLATILITY

Why it matters — the numerator of *ILLIQ* is the absolute daily return, which is mechanically larger whenever volatility is high, independent of trading volume or true price impact. A stock (or a market period) with high fundamental volatility but perfectly liquid, deep trading will show an elevated *ILLIQ* even though its price impact per dollar of volume has not changed, simply because the numerator moved. Potential bias — the paper's own SDRET control only weakly disentangles this (correlation of 0.278 between ILLIQ and SDRET, but this is a linear correlation across stock-years, not a decomposition of the ratio's variance into a volume-driven versus a volatility-driven component). How it could be tested — construct a measure that scales by expected (rather than realized) volatility, or use intraday range-based volatility estimators net of a separately estimated price-impact term, and check whether the time-series results survive.

PROBLEM: THE EQUAL-WEIGHTED MARKET RETURN CONCENTRATES THE RESULT IN SMALL STOCKS

Why it matters — because the market return used in equation (10) is equal-weighted, it is disproportionately influenced by the very small stocks that Table 3's own size-cut shows carry most of the illiquidity effect. The paper's headline market-level result may therefore be, to a meaningful extent, mechanically inherited from the small-stock illiquidity effect rather than an independent aggregate-market phenomenon. Potential bias — overstatement of how much a value-weighted, capitalization-representative "market" portfolio is exposed to time-varying illiquidity pricing. How it could be tested — re-estimate equation (10) using the value-weighted NYSE index as the dependent variable; the paper does not report this specification, only the equal-weighted index and the separate size-decile portfolios.

PROBLEM: REVERSE CAUSALITY BETWEEN VOLUME, RETURN, AND EXPECTATIONS

Why it matters — the AR(1) forecasting model treats illiquidity as the sole predetermined driver of expected return, but trading volume itself responds to information flow and to expected future returns (investors trade more when they revise expectations), so the direction of causality between illiquidity, volume, and subsequent returns is not pinned down by a bivariate autoregressive-plus-innovation regression. Potential bias — the estimated g_1, g_2 could partly reflect a common driver (e.g., aggregate uncertainty) that moves both trading activity and required returns, rather than a pure illiquidity-price channel. How it could be tested — a structural VAR relating volume, volatility, and returns, or an instrumental-variables approach using a liquidity shock plausibly orthogonal to fundamentals (e.g., regulatory changes in tick size or trading rules).

Single most important weakness. The paper cannot separate a genuine time-varying illiquidity premium from a time-varying volatility (or more general risk) premium, because its illiquidity measure is constructed from the same absolute-return numerator that any volatility-based risk-premium story would also predict to be priced. The size-decile monotonicity (Section 7.2) is the strongest available evidence against a pure risk-premium reinterpretation, since a naive risk story would need to separately explain why the same monotonic size gradient shows up for a ratio-based illiquidity measure and not simply for realized volatility itself — but the paper does not run the head-to-head horse race (illiquidity innovation versus volatility innovation, both included in equation 10) that would settle the question directly.

8. Contradictions and Edge Cases

When should the illiquidity effect disappear? If markets became so liquid that the cross-sectional dispersion in *ILLIQ* collapsed toward zero — broadly what happened to U.S. large-cap equities after the 2001 decimalization of tick sizes and the subsequent rise of high-frequency market making — the model in equation (10) predicts the estimated g_1 and g_2 coefficients should shrink toward zero, since there would be little cross-sectional or time-series variation left for the illiquidity variable to explain. EXTENSION This is directly testable out of sample by re-estimating equations (7) and (10) on post-1997 or post-2001 U.S. data; the reviewer is not aware of the paper itself, published in 2002, providing any such out-of-sample test, since the sample necessarily ends in 1997.

What happens at extreme values of illiquidity? The 1% trimming of the ILLIQ distribution in both tails (Section 4.2) is precisely designed to prevent the paper's results from being driven by extreme illiquidity events (e.g., a thinly traded stock with a single large one-day price move on negligible volume, which would produce an enormous, economically meaningless ILLIQ value). This is a reasonable precaution, but it also means the paper's own design explicitly excludes the very tail episodes — market crashes, flash-crash-type events — where illiquidity effects are, by the author's own account in the discussion of the October 1987 crash (Amihud, Mendelson and Wood, 1990), economically largest. The time-series results, in other words, describe "ordinary" fluctuations in aggregate illiquidity, not tail events.

Could another mechanism generate the same size gradient? Yes, in principle. A pure time-varying-risk-premium story (small stocks are more exposed to aggregate risk and its price rises when uncertainty rises) would generate exactly the same qualitative pattern — small-stock excess returns more sensitive to the state variable than large-stock excess returns — without any genuine liquidity channel at all, since aggregate uncertainty and aggregate illiquidity are themselves plausibly correlated (illiquidity tends to rise in market downturns and periods of high uncertainty). The paper's own citation of the default-premium coefficient's decline with size (Section 5.5) is, ironically, evidence for exactly this alternative story running in parallel.

Does the mechanism survive different samples? The paper's own subperiod splits (1964–1980 vs. 1981–1997 for the cross-section; six 68-month subperiods for the monthly time series) show the sign of the effect is stable, but the magnitude is not: the cross-sectional *ILLIQMA* coefficient falls from 0.135 in 1964–1980 to 0.088 in 1981–1997, roughly a one-third decline. INFERENCE This is consistent with, though not proof of, a secular decline in the illiquidity premium as U.S. equity market liquidity structurally improved over the sample (electronic order matching, narrower tick sizes toward the end of the period, rising institutional participation), a trend the paper does not explicitly discuss or extrapolate.

9. Further Literature

PRECEDED IT — THEORETICAL FOUNDATION

Amihud and Mendelson (1986), "Asset Pricing and the Bid-Ask Spread." Shared idea — the foundational claim that illiquidity, proxied by the bid-ask spread, commands a positive expected-return premium in the cross-section. Relationship — this 2002 paper is explicitly framed as extending Amihud and Mendelson's cross-sectional result into the time series using a different, volume-based proxy. Why it matters — without this prior result, the cross-sectional half of the 2002 paper would have no theoretical anchor; the entire time-series exercise is a natural next question once the cross-sectional relationship is taken as established.

Kyle (1985), "Continuous Auctions and Insider Trading." Shared idea — price impact as the market's response to order-flow imbalance under asymmetric information. Relationship — ILLIQ is explicitly presented as a coarse empirical analogue of Kyle's lambda. Why it matters — Kyle provides the microstructure logic that makes "absolute return per dollar of volume" a theoretically motivated quantity rather than an arbitrary ratio.

DEVELOPED SIMILAR IDEAS, DIFFERENT METHODOLOGY

Brennan and Subrahmanyam (1996), "Market Microstructure and Asset Pricing." Shared idea — illiquidity, measured from intraday transactions and quotes via Glosten-Harris estimation, is priced cross-sectionally. Relationship — this is the direct source of the 1984 microstructure benchmark used in Amihud (2002) to validate ILLIQ. Why it matters — it demonstrates the precision/availability tradeoff explicitly: Brennan-Subrahmanyam's measure is finer but exists only where intraday data exist, motivating Amihud's coarser but longer-horizon alternative.

APPEARED LATER, DEVELOPED THE CONTRIBUTION

Pastor and Stambaugh (2003), "Liquidity Risk and Expected Stock Returns," Journal of Political Economy. Shared idea — systematic, market-wide liquidity as a priced risk factor, tested via stocks' sensitivities (betas) to aggregate liquidity innovations rather than via a level characteristic. Relationship — this is the natural methodological successor: it reframes the "is illiquidity priced over time" question that Amihud (2002) answers via a predictive-return regression as a proper factor-model asset-pricing test with a traded liquidity-mimicking portfolio, addressing part of the identification weakness discussed in Section 7 (that a level-predictability regression cannot cleanly separate a price-of-risk story from a return-predictability story). Why it matters — it is arguably the paper's most direct and most cited intellectual heir. Amihud's 2002 working-paper version (cited within this article as Pastor and Stambaugh, 2001) predates the published 2003 version.

Acharya and Pedersen (2005), "Asset Pricing with Liquidity Risk," Journal of Financial Economics. Shared idea — liquidity risk enters expected returns through multiple covariance channels (level of illiquidity, and covariances of illiquidity with market return and with the asset's own return). Relationship — provides the fully specified equilibrium (liquidity-adjusted CAPM) that Amihud (2002) lacks; Amihud's paper is reduced-form and empirical, Acharya-Pedersen derive the pricing equation from an overlapping-generations equilibrium with transaction costs. Why it matters — it formalizes exactly the kind of theoretical mechanism that Section 3 of this review notes is only informally sketched in the 2002 paper.

Chordia, Roll and Subrahmanyam (2000, 2001), studies of commonality and time-series dynamics in liquidity. Shared idea — market-wide co-movement in liquidity and its relation to returns and order flow. Relationship — complements Amihud's aggregate ILLIQ series by documenting the microstructure sources (order imbalance, inventory risk) of aggregate liquidity commonality that Amihud's paper takes as a reduced-form, unmodeled aggregate. Why it matters — helps explain why a market-wide illiquidity aggregate is a meaningful object to begin with, rather than an average of idiosyncratic, uncorrelated stock-level frictions.

CONTRADICTS OR QUALIFIES ITS FINDINGS

Studies questioning the robustness of illiquidity-return relationships to microstructure-induced measurement artifacts (broadly, the concerns raised by Berk, 1995 and subsequent methodological critiques of characteristics that mechanically correlate with the deflator used to compute expected return). Relationship — Berk's critique, cited in the paper itself regarding size, applies with comparable force to any characteristic-based illiquidity measure constructed, as ILLIQ is, from a ratio involving price-related quantities; the paper acknowledges the analogous size critique but does not extend the same scrutiny explicitly to ILLIQ itself. Why it matters — it is a standing methodological qualification on the entire characteristics-based cross-sectional asset-pricing literature, of which this paper's Section 2 is a member.

10. Research Gaps

10.1 Paper-Level Gaps

- No head-to-head test of illiquidity innovations against volatility innovations in the same regression, leaving the volatility-versus-illiquidity confound (Section 7.3) unresolved.
- No value-weighted-market specification of the core time-series regression is reported, only equal-weighted and size-decile portfolios, leaving open how much of the aggregate result is attributable to small stocks mechanically dominating the equal-weighted index.
- The Chow stability test and the WLS and bias-simulation robustness checks are referenced but not reported in the published text, limiting independent verification.

10.2 Literature-Level Gaps

- At the time of writing, no fully specified equilibrium asset-pricing model connected a level-based illiquidity characteristic to a time-varying market-wide risk premium in a way that separately identified the price of illiquidity level from the price of illiquidity risk (covariance) — a gap subsequently addressed, in different ways, by Pastor and Stambaugh (2003) and Acharya and Pedersen (2005).
- Long time-series illiquidity measures for non-U.S. and emerging markets, where microstructure data are especially scarce and where the ILLIQ construction is potentially most useful, were largely unexplored at the time of publication.

10.3 Methodological Gaps

- **Data —** intraday data now exist for far longer spans than in 2002 (TAQ data extend from 1993, and many exchanges now provide full order-book history), which would allow a genuine head-to-head test of ILLIQ against microstructure-based measures across many years rather than the single-year (1984) benchmark used in the paper.
- **Identification —** a liquidity shock plausibly exogenous to fundamentals (regulatory tick-size changes, exchange-mandated market-maker obligations, circuit-breaker rule changes) could be used as an instrument or natural experiment to separate the illiquidity channel from confounding risk channels.
- **Estimation —** modern predictive-regression bias corrections (e.g., Lewellen 2004; Campbell and Yogo 2006) for persistent, endogenous regressors postdate this paper and would allow a sharper, better-powered re-estimation of equations (9)–(11) than the Kendall (1954) and Stambaugh (1999) corrections available to the original author.

11. Research Opportunities

1. A volatility-orthogonalized illiquidity measure and a horse race against ILLIQ

RESEARCH QUESTION Does the ILLIQ time-series predictability result survive when illiquidity innovations are orthogonalized against contemporaneous realized-volatility innovations?

MOTIVATION Section 7.3 identifies the volatility-illiquidity numerator confound as the paper's central unresolved weakness.

HYPOTHESIS A meaningful fraction, but not all, of the predictive power of AILLIQ for market excess return is subsumed by realized-volatility innovations.

ECONOMIC MECHANISM If price impact and volatility are jointly driven by information flow, decomposing their common component isolates the pure liquidity-price channel from the risk-price channel.

LITERATURE CONNECTION Extends Amihud (2002) directly; relates to Pastor and Stambaugh (2003) liquidity-beta approach and to volatility-risk-premium studies (French, Schwert and Stambaugh, 1987).

DATA CRSP daily returns and volume, 1963–present, extending the original sample by two and a half decades.

VARIABLES AILLIQ, an intraday-range or realized-volatility estimator, and their orthogonalized innovations from a bivariate VAR.

METHOD Bivariate VAR(1) in log illiquidity and log realized volatility; predictive regression of market excess return on both orthogonalized innovations.

IDENTIFICATION STRATEGY Orthogonalization via Cholesky decomposition of the VAR residual covariance, with results checked for ordering sensitivity.

EXPECTED CONTRIBUTION A cleaner attribution of how much of the historical equity premium's time variation is genuinely a liquidity premium versus a volatility/risk premium.

MAIN RISK Illiquidity and volatility may be so tightly intertwined mechanically (via the shared absolute-return numerator) that orthogonalization is not economically meaningful, only a statistical relabeling.

2. Out-of-sample re-estimation through the decimalization and high-frequency-trading eras

RESEARCH QUESTION Has the illiquidity premium documented for 1963–1997 weakened, strengthened, or changed shape since decimalization (2001) and the rise of algorithmic market making?

MOTIVATION Section 8 notes the paper's own subperiod split shows a declining cross-sectional coefficient into the 1980s and 1990s; the post-2001 microstructure regime is a genuinely new environment the original paper could not observe.

HYPOTHESIS The level of the illiquidity premium has declined further post-2001, but the size-gradient (SZ-1, SZ-2) persists because small-cap liquidity provision remains comparatively costly even in an electronic market.

ECONOMIC MECHANISM Narrower tick sizes and algorithmic market making compress bid-ask spreads and price impact for large, liquid names disproportionately more than for small, thinly traded names.

LITERATURE CONNECTION Direct replication-extension of Amihud (2002); relates to market-quality studies of decimalization (e.g., Bessembinder, 2003).

DATA CRSP daily data 1998–2025, re-estimating equations (1), (3), (4), (7) and (10) on the post-sample period.

VARIABLES Identical to the original paper, for direct comparability.

METHOD Exact replication of the paper's Fama-MacBeth and AR(1)-plus-innovation methodology on the extended sample, with a formal structural-break test at 2001.

IDENTIFICATION STRATEGY Decimalization as a regulatory event providing a natural pre/post break rather than an arbitrary sample split.

EXPECTED CONTRIBUTION A direct, up-to-date test of whether the paper's central time-series claim still holds a quarter-century later, and whether its magnitude has changed in the predicted direction.

MAIN RISK Structural changes in trading technology may have altered the meaning of "dollar volume" itself (e.g., high-frequency round-trip volume inflates VOLD without corresponding liquidity for large orders), requiring a redefinition of ILLIQ rather than a simple re-estimation.

3. A liquidity-adjusted CAPM test using ILLIQ-style measures in markets without transaction-level data

RESEARCH QUESTION Does the Acharya and Pedersen (2005) liquidity-adjusted CAPM framework, when its liquidity-covariance terms are built from Amihud-style ILLIQ rather than transaction-level data, produce a priced illiquidity-risk factor in emerging or frontier equity markets where microstructure data do not exist?

MOTIVATION The explicit rationale for ILLIQ—availability where finer measures are absent—is most valuable precisely in markets that lack transaction-level data, which the original paper (NYSE only) does not test.

HYPOTHESIS Liquidity-risk premia estimated via ILLIQ-based covariances in less liquid, less researched equity markets are larger in magnitude than the NYSE estimates in Amihud (2002), reflecting a scarcer supply of liquidity provision.

ECONOMIC MECHANISM Fewer market makers and lower institutional participation in smaller markets should raise the equilibrium compensation demanded for bearing illiquidity risk.

LITERATURE CONNECTION Combines Amihud (2002)'s measurement approach with Acharya and Pedersen (2005)'s equilibrium pricing framework; connects to emerging-market liquidity studies (e.g., Bekaert, Harvey and Lundblad, 2007).

DATA Daily return and volume data from a sample of emerging-market exchanges (e.g., Datastream or Refinitiv, or national exchange archives), ideally 15+ years per market for adequate AR(1) estimation.

VARIABLES Market-level and stock-level ILLIQ, its covariances with market return and own-stock return, and standard risk-factor controls.

METHOD Acharya-Pedersen's liquidity-adjusted CAPM regressions estimated with ILLIQ-based covariance proxies in place of the original transaction-cost-based measures.

IDENTIFICATION STRATEGY Cross-market comparison (NYSE benchmark versus emerging markets) holding the measurement methodology fixed, isolating market-structure differences as the source of any premium difference.

EXPECTED CONTRIBUTION Would extend the paper's explicit design goal, availability in data-poor markets, to the equilibrium asset-pricing framework it was never tested against in its own setting.

MAIN RISK Cross-country institutional differences (foreign ownership limits, capital controls, thin markets with extreme outliers) could dominate any genuine liquidity-risk signal, requiring careful outlier and regime controls beyond the original paper's 1% trimming rule.

4. Decomposing the size premium's time variation into illiquidity-level and illiquidity-risk components

RESEARCH QUESTION How much of the year-to-year variation in the realized small-minus-large size premium is statistically attributable to variation in expected and unexpected market illiquidity, as opposed to other candidate drivers (default risk, momentum crowding, macro state variables)?

MOTIVATION Section 7.1 notes that the paper's size-effect claim is suggestive, not quantified; this idea directly quantifies it.

HYPOTHESIS Expected and unexpected illiquidity jointly explain a statistically and economically significant, but partial (well under half), share of the time-series variance of the realized size premium.

ECONOMIC MECHANISM As in the paper's own logic, the size premium is partly a compensation for the illiquidity premium's disproportionate loading on small stocks (Table 3's SZ-1/SZ-2 pattern).

LITERATURE CONNECTION Builds on Amihud (2002) and Brown, Kleidon and Marsh (1983)'s documentation of size-premium instability; relates to Fama and French (1993) factor-based decompositions.

DATA CRSP size-decile portfolio returns 1964–present; AILLIQ series extended per idea 2; default and term premium series from FRED.

VARIABLES RSZ2–RSZ10 spread return; lagged and unexpected AILLIQ; DEF; TERM; a business-cycle indicator.

METHOD Variance decomposition (e.g., an R^2-based Shapley or LMG decomposition) of the size-premium spread regression on the candidate drivers.

IDENTIFICATION STRATEGY Purely a predictive-decomposition exercise; causal claims would require the instrument discussed in idea 1 or a natural experiment.

EXPECTED CONTRIBUTION A quantified, rather than qualitative, answer to how much of the size anomaly's notorious instability is an illiquidity story.

MAIN RISK Variance decompositions are sensitive to variable ordering and to omitted correlated drivers; results could be fragile to specification choices.

5. A market-crash-focused re-estimation using non-trimmed tail observations

RESEARCH QUESTION Does the expected/unexpected illiquidity mechanism (equations 7–10) hold, and if so with what magnitude, during and immediately after major liquidity-crisis episodes (1987 crash, 1998 LTCM, 2008 financial crisis, March 2020) that the original paper's 1% trimming rule explicitly excludes from its ordinary-sample estimates?

MOTIVATION Section 8 notes the paper's design excludes exactly the tail episodes where its own cited prior work (Amihud, Mendelson and Wood, 1990, on 1987) suggests illiquidity effects are largest.

HYPOTHESIS The coefficients g1 and g2, estimated on windows including crisis episodes without trimming, are substantially larger in magnitude than the full-sample trimmed estimates, and the size-gradient (SZ-1, SZ-2) steepens further during crises.

ECONOMIC MECHANISM Flight-to-liquidity dynamics (explicitly discussed by the author regarding 1987) should be strongest exactly when illiquidity itself spikes to levels the ordinary-sample trimming rule discards.

LITERATURE CONNECTION Extends the crash-specific analysis in Amihud, Mendelson and Wood (1990) using the ILLIQ framework of Amihud (2002); connects to the broader flight-to-liquidity and liquidity-spiral literature (e.g., Brunnermeier and Pedersen, 2009).

DATA High-frequency or daily CRSP data around each crisis window, deliberately retaining tail ILLIQ observations rather than trimming them.

VARIABLES Daily or weekly ILLIQ and AILLIQ around each event window; size-decile portfolio returns.

METHOD Event-study regressions of the type used in crash studies, combined with the paper's own AR(1)-plus-innovation structure adapted to a higher (weekly or daily) frequency appropriate for crisis dynamics.

IDENTIFICATION STRATEGY Event-window identification around exogenous crisis triggers, rather than the paper's calendar-time predictive-regression design.

EXPECTED CONTRIBUTION Would test whether the paper's "ordinary times" mechanism extrapolates to the tail episodes it was never designed to cover, directly addressing the edge case raised in Section 8.

MAIN RISK Crisis episodes are few and idiosyncratic (small effective sample of events), and untrimmed extreme ILLIQ values may be dominated by data errors or extreme illiquidity in a handful of near-halted names rather than a systematic market-wide phenomenon.

Ranking of the Strongest Ideas

Ranked by novelty, feasibility, data availability, methodological quality, economic importance and potential contribution, the top three ideas are, in order: **Idea 2** (out-of-sample re-estimation through decimalization and HFT eras), because it is the most directly feasible with freely available CRSP data, requires no new theory, and answers the most pressing question a contemporary reader has about a 2002 paper — does the result still hold; **Idea 1** (volatility-orthogonalized illiquidity), because it targets the single most important identified weakness of the paper (Section 7.3) with a clean, implementable econometric design; and **Idea 4** (variance decomposition of the size premium), because it converts the paper's most cited but least quantified claim into a testable, quantified contribution using data that are already public. Ideas 3 and 5 are economically important but carry materially higher data-acquisition and small-sample risk.

12. Learning Framework

Prerequisite Concepts

- The Fama-MacBeth (1973) two-step cross-sectional estimation procedure and why its standard errors are computed from the time series of monthly coefficients rather than from a single pooled panel regression.
- Basic time-series concepts: autoregressive (AR(1)) processes, persistence, stationarity, and the small-sample downward bias in OLS-estimated AR coefficients.
- The concept of price impact in market microstructure, particularly Kyle's (1985) lambda.
- Predictive-regression bias in finance (Stambaugh, 1999): why regressing returns on a lagged, persistent, endogenous predictor produces biased coefficients.

Core Concepts Learned

- How to construct a volume-based illiquidity proxy from data that virtually every equity market publishes, and why this measure trades precision for availability relative to bid-ask-spread or transactions-based measures.
- How to decompose a return-predicting variable into an expected (forecastable) and unexpected (innovation) component using a first-stage AR(1) model, and use both components in a second-stage return regression.
- How a cross-sectional characteristic-pricing result (illiquidity is priced across stocks) can be extended, with appropriate care, into a time-series predictability claim (illiquidity is priced across time).

Important Equations

- Equation (1): the ILLIQ measure itself — absolute return per dollar of volume, averaged over a period.
- Equations (7)–(9): the AR(1) expectation-formation model and its substitution into the return equation, the paper's central identification device.
- Equation (10): the estimated reduced-form regression combining expected and unexpected illiquidity, the workhorse specification of the entire empirical section.

Methods Learned

- Fama-MacBeth cross-sectional estimation with characteristic-based portfolio betas (Scholes-Williams method).

- Kendall's (1954) small-sample bias correction for AR(1) coefficients.
- Newey-West (1987) and White (1980) heteroskedasticity- and autocorrelation-consistent standard errors in a predictive-regression context.

Questions to Test Understanding

COMPREHENSION

- Explain, without equations, why the paper needs an AR(1) model of illiquidity before it can test whether "expected illiquidity" predicts returns.
- What is the economic difference between the effect of expected illiquidity (H-1) and unexpected illiquidity (H-2) on stock returns?

ANALYSIS

- Why does substituting equation (8) into equation (6) make g_1 a product of the illiquidity price φ_1 and the illiquidity persistence c_1, and why does this matter for interpreting the size of g_1?
- Why might the equal-weighted market return used in the time-series tests overstate the aggregate importance of the illiquidity channel relative to a value-weighted index?

RESEARCH

- Design a test that could distinguish a genuine illiquidity-price effect from a volatility-price effect using only the variables already available in the paper's dataset.
- If you re-estimated this paper's equations on 1998–2025 U.S. data, what specific change in the market's trading technology would you expect to alter the definition, not just the estimated value, of ILLIQ?

Papers to Read Next

- Pastor, L., Stambaugh, R.F. (2003). Liquidity Risk and Expected Stock Returns. *Journal of Political Economy*, 111(3), 642–685.
- Acharya, V.V., Pedersen, L.H. (2005). Asset Pricing with Liquidity Risk. *Journal of Financial Economics*, 77(2), 375–410.
- Chordia, T., Subrahmanyam, A., Anshuman, V.R. (2001). Trading Activity and Expected Stock Returns. *Journal of Financial Economics*, 59(1), 3–32.

13. Conclusion

Amihud (2002) makes a durable methodological contribution — a volume-based illiquidity proxy usable wherever ordinary daily return and volume data exist — and uses it to extend a well-established cross-sectional result (illiquid stocks earn higher average returns) into a genuinely new time-series claim (expected market-wide illiquidity forecasts the market excess return, and unexpected illiquidity moves prices contemporaneously). The size-decile cross-cut, showing both effects concentrated in small stocks, is the paper's most persuasive piece of internal evidence, since it is difficult to generate that specific monotonic pattern from a purely spurious correlation. At the same time, the paper's central limitation, that its illiquidity measure cannot be cleanly separated from return volatility, and that its time-series tests rest on a small number of highly persistent annual observations, means the paper is best read as establishing a robust, economically important stylized fact rather than a fully identified structural mechanism. Its lasting influence is visible less in its own specific coefficients than in the research program it opened, most directly realized in Pastor and Stambaugh (2003) and Acharya and Pedersen (2005), which built the equilibrium liquidity-risk-pricing theory this paper's reduced-form evidence called for but did not itself supply.

One-Sentence Thesis

Expected market-wide illiquidity, measured by a simple volume-scaled absolute-return ratio, positively predicts the market's ex ante excess return, while unexpected illiquidity depresses contemporaneous returns, with both effects strongest for small stocks.

Three Most Important Insights

- A coarse, universally available illiquidity proxy can substitute for microstructure data in time-series asset-pricing tests that transaction-level measures cannot support for lack of sufficiently long samples.
- The equity premium can be decomposed, at least in part, into a risk component and an illiquidity component, providing one candidate explanation for why the historical equity premium looks large relative to standard risk-based models.
- Time variation in the small-firm premium is linked to time variation in aggregate market illiquidity, connecting two previously separate anomaly literatures.

Strongest Evidence

The monotonic decline in the illiquidity coefficients across size deciles (Table 3 and Table 4, hypotheses SZ-1 and SZ-2), replicated independently at both annual and monthly frequency and surviving the addition of bond-yield-premium controls.

Biggest Weakness

The inability to separate a genuine price-impact-based illiquidity channel from a volatility-based risk channel, since the paper's own illiquidity measure is constructed from the absolute daily return, the same numerator that any pure risk story would also predict to matter.

Most Important Research Gap

No published test in the paper directly decomposes the illiquidity and volatility channels against each other in the same regression, leaving the paper's central causal interpretation suggestive rather than demonstrated.

Best Research Opportunity

Out-of-sample re-estimation of the paper's exact methodology through the decimalization and algorithmic-trading era (Research Idea 2), which is immediately feasible with public CRSP data and would show directly whether the time-series illiquidity premium the paper documents for 1963–1997 has persisted, weakened, or vanished.

Three Papers to Read Next

- Pastor, L., Stambaugh, R.F. (2003). Liquidity Risk and Expected Stock Returns. *Journal of Political Economy*, 111(3), 642–685.
- Acharya, V.V., Pedersen, L.H. (2005). Asset Pricing with Liquidity Risk. *Journal of Financial Economics*, 77(2), 375–410.
- Amihud, Y., Mendelson, H. (1986). Asset Pricing and the Bid-Ask Spread. *Journal of Financial Economics*, 17(2), 223–249.

14. References

Acharya, V.V., Pedersen, L.H., 2005. Asset pricing with liquidity risk. Journal of Financial Economics 77, 375–410.

Amihud, Y., Mendelson, H., 1980. Dealership market: market making with inventory. Journal of Financial Economics 8, 311–353.

Amihud, Y., Mendelson, H., 1986. Asset pricing and the bid-ask spread. Journal of Financial Economics 17, 223–249.

Amihud, Y., Mendelson, H., 1989. The effects of beta, bid-ask spread, residual risk and size on stock returns. Journal of Finance 44, 479–486.

Amihud, Y., Mendelson, H., Wood, R., 1990. Liquidity and the 1987 stock market crash. Journal of Portfolio Management 16, 65–69.

Amihud, Y., Mendelson, H., 1991a. Liquidity, maturity and the yields on U.S. government securities. Journal of Finance 46, 1411–1426.

Amihud, Y., Mendelson, H., 1991b. Liquidity, asset prices and financial policy. Financial Analysts Journal 47, 56–66.

Amihud, Y., Mendelson, H., Lauterbach, B., 1997. Market microstructure and securities values: evidence from the Tel Aviv exchange. Journal of Financial Economics 45, 365–390.

Amihud, Y., 2002. Illiquidity and stock returns: cross-section and time-series effects. Journal of Financial Markets 5, 31–56.

Atkins, A.B., Dyl, E.A., 1997. Transactions costs and holding periods for common stocks. Journal of Finance 52, 309–325.

Banz, R.W., 1981. The relationship between return and market value of common stocks. Journal of Financial Economics 9, 3–18.

Barry, C.B., Brown, S.J., 1984. Differential information and the small firm effect. Journal of Financial Economics 13, 283–294.

Berk, J., 1995. A critique of size-related anomalies. Review of Financial Studies 8, 275–286.

Berkman, H., Eleswarapu, V.R., 1998. Short-term traders and liquidity: a test using Bombay stock exchange data. Journal of Financial Economics 47, 339–355.

Boudoukh, J., Richardson, M., Smith, T., 1993. Is the ex ante risk premium always positive? Journal of Financial Economics 34, 387–408.

Brennan, M.J., Subrahmanyam, A., 1996. Market microstructure and asset pricing: on the compensation for illiquidity in stock returns. Journal of Financial Economics 41, 441–464.

Brennan, M.J., Chordia, T., Subrahmanyam, A., 1998. Alternative factor specifications, security characteristics, and the cross-section of expected stock returns. Journal of Financial Economics 49, 345–373.

Brown, P., Kleidon, A.W., Marsh, T.A., 1983. New evidence on the nature of size-related anomalies in stock prices. Journal of Financial Economics 12, 33–56.

Brunnermeier, M.K., Pedersen, L.H., 2009. Market liquidity and funding liquidity. Review of Financial Studies 22, 2201–2238.

Chakravarty, S., Sarkar, A., 1999. Liquidity in U.S. fixed income markets: a comparison of the bid-ask spread in corporate, government and municipal bond markets. Federal Reserve Bank of New York, Staff Report 73.

Chalmers, J.M.R., Kadlec, G.B., 1998. An empirical examination of the amortized spread. Journal of Financial Economics 48, 159–188.

Chordia, T., Roll, R., Subrahmanyam, A., 2000. Commonality in liquidity. Journal of Financial Economics 56, 3–28.

Chordia, T., Subrahmanyam, A., Anshuman, V.R., 2001. Trading activity and expected stock returns. Journal of Financial Economics 59, 3–32.

Constantinides, G.M., Scholes, M.S., 1980. Optimal liquidation of assets in the presence of personal taxes: implications for asset pricing. Journal of Finance 35, 439–443.

Constantinides, G.M., 1986. Capital market equilibrium with transaction costs. Journal of Political Economy 94, 842–862.

Cooper, S.K., Groth, J.C., Avera, W.E., 1985. Liquidity, exchange listing and common stock performance. Journal of Economics and Business 37, 19–33.

Copeland, T.C., Galai, D., 1983. Information effect on the bid-ask spread. Journal of Finance 38, 1457–1469.

Datar, V.T., Naik, N.Y., Radcliffe, R., 1998. Liquidity and stock returns: an alternative test. Journal of Financial Markets 1, 205–219.

Easley, D., O'Hara, M., 1987. Price, trade size and information in securities markets. Journal of Financial Economics 19, 69–90.

Easley, D., Hvidkjaer, S., O'Hara, M., 1999. Is information risk a determinant of asset returns? Working paper, Cornell University.

Eleswarapu, V.R., Reinganum, M., 1993. The seasonal behavior of liquidity premium in asset pricing. Journal of Financial Economics 34, 373–386.

Eleswarapu, V.R., 1997. Cost of transacting and expected returns in the Nasdaq market. Journal of Finance 52, 2113–2127.

Fama, E.F., MacBeth, J.D., 1973. Risk, return and equilibrium: empirical tests. Journal of Political Economy 81, 607–636.

Fama, E.F., French, K.R., 1989. Business conditions and expected returns on stocks and bonds. Journal of Financial Economics 25, 23–49.

Fama, E.F., 1990. Stock returns, expected returns, and real activity. Journal of Finance 45, 1089–1108.

Fama, E.F., French, K.R., 1992. The cross-section of expected stock returns. Journal of Finance 47, 427–465.

Fama, E.F., French, K.R., 1993. Common risk factors in the returns on stocks and bonds. Journal of Financial Economics 33, 3–56.

French, K.R., Schwert, G.W., Stambaugh, R.F., 1987. Expected stock returns and volatility. Journal of Financial Economics 19, 3–29.

Glosten, L.R., Milgrom, P.R., 1985. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. Journal of Financial Economics 14, 71–100.

Glosten, L., Harris, L., 1988. Estimating the components of the bid-ask spread. Journal of Financial Economics 21, 123–142.

Harris, M., Raviv, A., 1993. Differences of opinion make a horse race. Review of Financial Studies 6, 473–506.

Harris, L.E., 1994. Minimum price variation, discrete bid-ask spreads, and quotation sizes. Review of Financial Studies 7, 149–178.

Hasbrouck, J., 1991. Measuring the information content of stock trades. Journal of Finance 46, 179–207.

Haugen, R.A., Baker, N.L., 1996. Commonality in the determinants of expected stock returns. Journal of Financial Economics 41, 401–439.

Kamara, A., 1994. Liquidity, taxes, and short-term treasury yields. Journal of Financial and Quantitative Analysis 29, 403–416.

Keim, D.B., 1983. Size related anomalies and stock return seasonality. Journal of Financial Economics 12, 13–32.

Keim, D.B., Stambaugh, R.F., 1986. Predicting returns in the stock and bond market. Journal of Financial Economics 17, 357–396.

Keim, D.B., Madhavan, A., 1996. The upstairs market for large-block transactions: analysis and measurement of price effects. Review of Financial Studies 9, 1–36.

Kendall, M.G., 1954. Note on bias in the estimation of autocorrelation. Biometrika 41, 403–404.

Kraus, A., Stoll, H.R., 1972. Price impacts of block trading on the New York Stock Exchange. Journal of Finance 27, 569–588.

Kyle, A.S., 1985. Continuous auctions and insider trading. Econometrica 53, 1315–1335.

Levy, H., 1978. Equilibrium in an imperfect market: constraint on the number of securities in the portfolio. American Economic Review 68, 643–658.

Loughran, T., 1997. Book-to-market across firm size, exchange and seasonality: is there an effect? Journal of Financial and Quantitative Analysis 32, 249–268.

Merton, R.C., 1987. A simple model of capital market equilibrium with incomplete information. Journal of Finance 42, 483–511.

Newey, W.K., West, K.D., 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. Econometrica 55, 703–706.

Pastor, L., Stambaugh, R.F., 2003. Liquidity risk and expected stock returns. Journal of Political Economy 111, 642–685.

Redding, L.S., 1997. Firm size and dividend payouts. Journal of Financial Intermediation 6, 224–248.

Reinganum, M.R., 1981. Misspecification of capital asset pricing: empirical anomalies based on earnings' yields and market values. Journal of Financial Economics 9, 19–46.

Reinganum, M.R., 1990. Market microstructure and asset pricing. Journal of Financial Economics 28, 127–147.

Rouwenhorst, K.G., 1998. Local return factors and turnover in emerging stock markets. Working paper, Yale University.

Sawa, T., 1978. The exact moments of the least squares estimator for the autoregressive model. Journal of Econometrics 8, 159–172.

Scholes, M., Williams, J., 1977. Estimating betas from non-synchronous data. Journal of Financial Economics 5, 309–327.

Shumway, T., 1997. The delisting bias in CRSP data. Journal of Finance 52, 327–340.

Silber, W.L., 1975. Thinness in capital markets: the case of the Tel Aviv Stock Exchange. Journal of Financial and Quantitative Analysis 10, 129–142.

Stambaugh, R.F., 1999. Predictive regressions. Journal of Financial Economics 54, 375–421.

Stigum, M., 1983. The Money Market. Dow Jones-Irwin, Homewood, IL.

Stoll, H.R., 1978. The pricing of security dealer services: an empirical study of Nasdaq stocks. Journal of Finance 33, 1153–1172.

Stoll, H.R., Whaley, R.H., 1983. Transaction costs and the small firm effect. Journal of Financial Economics 12, 57–79.

Tinic, S.M., West, R.R., 1986. Risk, return and equilibrium: a revisit. Journal of Political Economy 94, 126–147.